ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 5

MATHEMATICS

PAPER 1 · PRACTICE A

MARKS: 60 **TIME: 1¼ hours**

An end-of-year test on Grade 5 Mathematics: whole numbers up to 6 digits, multiples and factors, number patterns and flow diagrams, the four operations, common fractions, angles, area and perimeter, length and mass, 3D objects, symmetry, time, a pictograph and probability.

Written by Academic Central for practice, following the CAPS curriculum for Grade 5. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL your working. You can get marks for working even if the answer is wrong.
4. You may NOT use a calculator.
5. Write neatly.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 What is the value of the digit 6 in 46 352?

A 6 B 600 C 6 000 D 60 000

1.2 300 000 + 40 000 + 500 + 8 = …

A 340 508 B 345 008 C 34 058 D 304 508

1.3 Which number is a multiple of 7?

A 27 B 35 C 45 D 57

1.4 The factors of 12 are …

A 1, 2, 3, 4, 6 and 12. B 2, 4, 6, 8, 10 and 12.
C 12, 24 and 36. D only 1 and 12.

1.5 An angle that is smaller than a right angle is …

A obtuse. B acute. C straight. D reflex.

1.6 Which fraction is equal to $\frac{2}{3}$?

A $\frac{4}{6}$ B $\frac{2}{6}$ C $\frac{3}{2}$ D $\frac{4}{3}$

1.7 How many days are there in a leap year?

A 364 B 365 C 366 D 360

1.8 You toss a coin once. Which sentence is true?

A It will certainly land on heads. B It cannot land on heads.
C Heads and tails are equally likely. D Tails is more likely than heads.

1.9 Which 3D object has one flat circular face, one curved surface and an apex (a point)?

A Cone B Cylinder C Cube D Pyramid

1.10 500 ml + 750 ml = …

A 1 ℓ 250 ml B 1 ℓ 200 ml
C 12 ℓ 50 ml D 1 250 ℓ

[10]

QUESTION 2: NUMBERS AND PATTERNS

2.1 Write 70 405 in words. (1)

2.2 Write 42 316 in expanded notation. (1)

2.3 Round 38 762 off to the nearest 1 000. (1)

2.4 Write <, > or = between the numbers: 56 789 ___ 56 798 (1)

2.5 Look at the number line in FIGURE 1. Which numbers are at A and at B? (2)

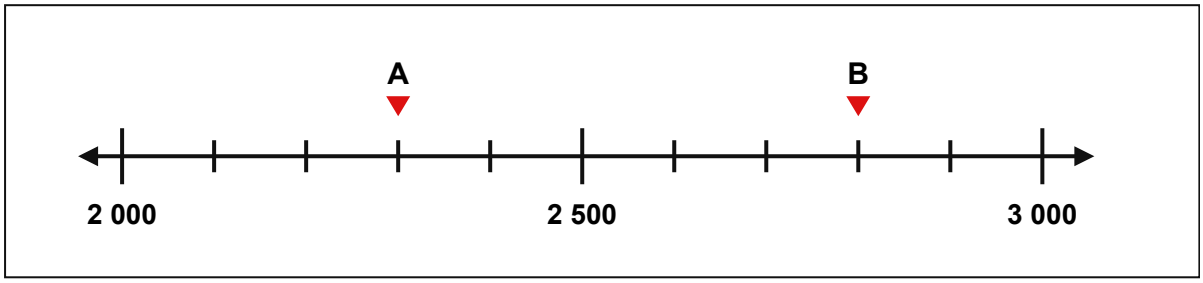


FIGURE 1: A number line

2.6 Look at the flow diagram. The rule is: multiply by 4, then add 1. Write the output for each input.

Input → **× 4** → **+ 1** → Output

2.6.1 Input 3 (1)

2.6.2 Input 5 (1)

2.6.3 Input 8 (1)

2.7 Write the next TWO numbers in the pattern: 4; 9; 14; 19; ___; ___ (2)

2.8 What number is 1 000 less than 60 250? (1)

[12]

QUESTION 3: CALCULATIONS

Show your working for each question.

3.1 24 567 + 13 845 (2)

3.2 50 000 − 17 384 (2)

3.3 236 × 24 (3)

3.4 864 ÷ 12 (2)

3.5 A farmer packs 875 eggs into trays. Each tray holds 30 eggs. How many trays can he fill completely, and how many eggs are left over? (3)

3.6 Thandi buys 3 books. Each book costs R125. How much does she pay? (2)

[14]

QUESTION 4: COMMON FRACTIONS

4.1 What fraction of the rectangle in FIGURE 2 is shaded? Write the fraction in its simplest form. (2)

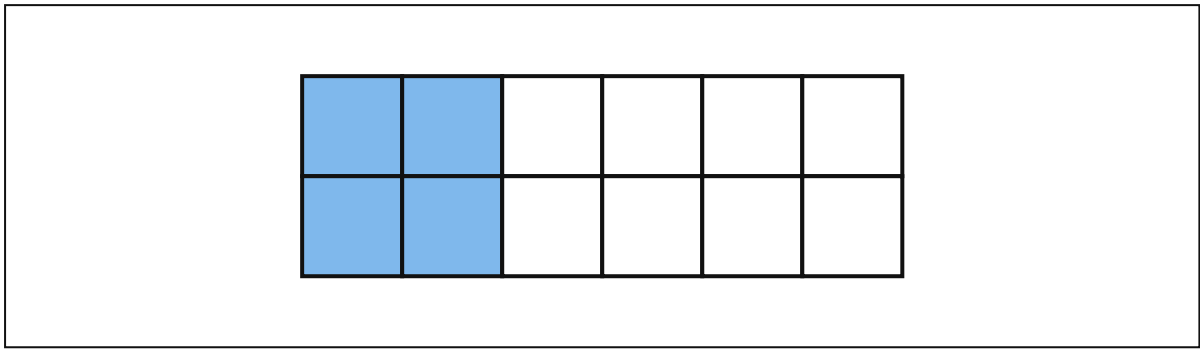


FIGURE 2: A rectangle in equal parts

4.2 $\frac{3}{8} + \frac{2}{8}$ (1)

4.3 $\frac{7}{10} - \frac{3}{10}$ (1)

4.4 Find $\frac{3}{5}$ of 40. (2)

4.5 Write these fractions from the smallest to the biggest: $\frac{1}{2}$; $\frac{1}{8}$; $\frac{1}{4}$ (1)

4.6 Write $\frac{7}{4}$ as a mixed number. (1)

[8]

QUESTION 5: SPACE, SHAPE AND MEASUREMENT

5.1 Write the type of each angle in FIGURE 3: acute, right or obtuse. (3)

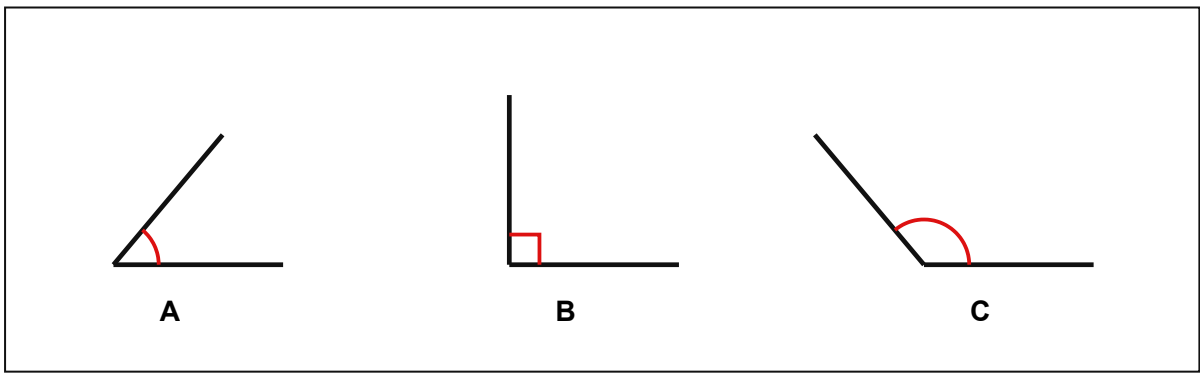


FIGURE 3: Three angles

5.2 FIGURE 4 shows a shape on a grid of 1 cm squares.

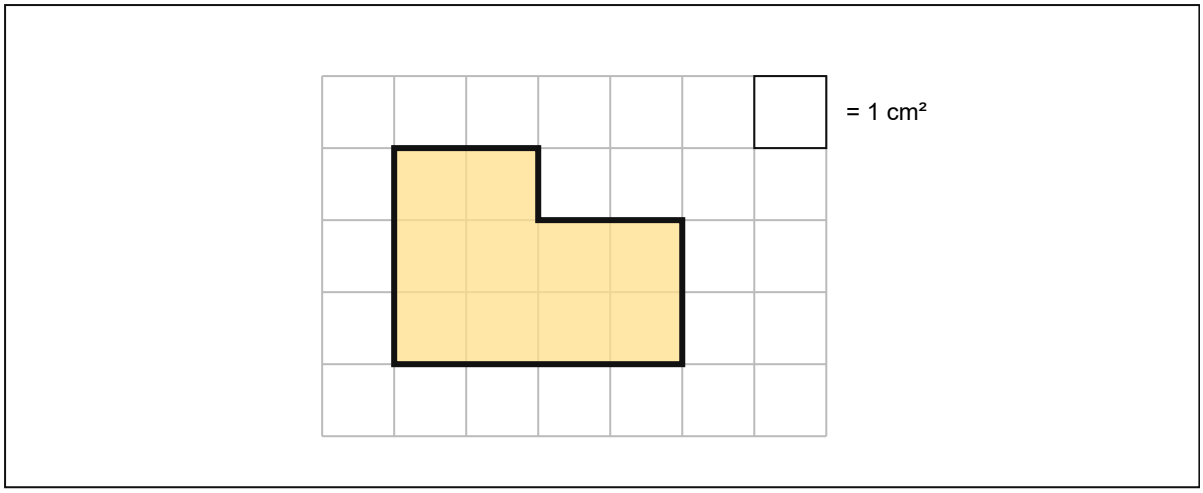


FIGURE 4: A shape on a grid

5.2.1 What is the area of the shape? (1)

5.2.2 What is the perimeter of the shape? Show how you worked it out. (2)

5.3 How many centimetres are there in 3 m? (1)

5.4 A box has a mass of 2 kg 400 g. Write its mass in grams. (1)

5.5 Name the 3D object that has the shape of a tin of beans. (1)

5.6 How many lines of symmetry does a rectangle have? (1)

[10]

QUESTION 6: TIME, DATA AND PROBABILITY

6.1 A movie starts at 14:35 and lasts 1 hour 50 minutes. At what time does it end? (2)

6.2 The pictograph in FIGURE 5 shows how many books four friends read in a month.

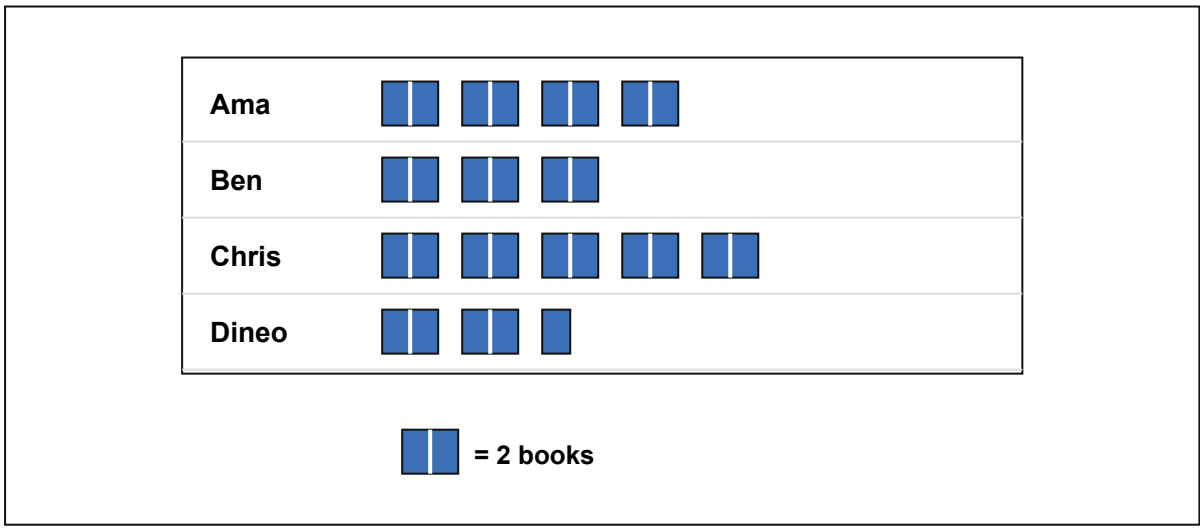


FIGURE 5: Books we read in a month

6.2.1 How many books did Chris read? (1)

6.2.2 How many more books did Ama read than Dineo? (1)

6.2.3 How many books did the four friends read altogether? (1)

6.3 A bag has 3 red balls and 1 blue ball. You take out one ball without looking. Which colour are you most likely to get? (1)

[6]

TOTAL: 60

END OF QUESTION PAPER